

PORT HEIDEN, AK, USA

WMO: 703330

Lat: 56.959N

Lon: 158.632W

Elev: 27

StdP: 101.00

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25508

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-17.7	-30.3	0.2	-10.6	-26.5	0.3	-7.7	19.2	3.3	17.5	3.7	6.8	50	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	18.2	13.8	17.1	12.8	15.8	12.2	14.8	17.7	13.6	16.1	12.7	14.9	6.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	9.5	15.9	12.4	9.0	15.0	11.3	8.4	13.9	41.3	17.9	38.1	16.1	35.9	15.0	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.2	12.2	DB	-23.8	21.8	3.6	2.0	-26.4	23.2	-28.5	24.4	-30.5	25.6	-33.1	27.1	(4)
(5)			WB	-23.7	16.7	3.5	1.5	-26.2	17.8	-28.3	18.6	-30.3	19.5	-32.8	20.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-4.2	-2.5	-3.3	0.6	4.9	8.4	10.9	11.7	9.2	4.3	0.5	-2.3
	DBStd	7.21	7.13	6.42	5.77	3.70	3.16	2.76	2.33	2.24	2.51	3.85	5.00	6.16
	HDD10.0	2613	440	349	413	281	159	59	17	7	43	178	286	382
	HDD18.3	5513	698	582	671	531	416	297	229	204	275	435	535	639
	CDD10.0	142	0	0	0	0	2	12	46	61	18	2	1	2
	CDD18.3	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDH23.3	7	0	0	0	0	0	0	0	0	0	0	0	7
	CDH26.7	3	0	0	0	0	0	0	0	0	0	0	0	3
Wind	WSAvg	5.8	6.1	6.5	6.1	5.8	5.4	5.2	4.8	5.2	5.7	5.9	6.2	6.5
Precipitation	PrecAvg	909	82	69	56	48	51	61	78	104	97	88	85	90
	PrecMax	1193	278	160	117	92	100	96	135	155	150	161	143	178
	PrecMin	698	20	12	9	12	11	33	30	56	48	32	29	32
	PrecStd	130	44	31	29	18	22	17	22	22	24	30	30	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	8.0	7.9	12.2	17.3	18.0	21.3	21.4	18.0	13.7	12.1	11.3
		MCWB	4.5	5.2	4.6	8.3	8.4	12.4	15.4	15.9	14.0	10.3	9.2	5.2
	2%	DB	6.3	7.0	6.3	8.7	13.0	16.3	17.7	18.2	15.9	12.3	9.9	7.6
		MCWB	3.4	4.4	3.3	5.5	7.9	10.8	13.5	14.5	12.4	9.8	7.9	4.8
	5%	DB	5.0	5.9	4.8	7.3	11.6	14.1	16.3	17.1	14.0	11.1	8.0	6.2
		MCWB	2.8	3.6	2.4	4.2	7.2	9.8	12.5	13.6	11.0	8.9	5.8	3.6
	10%	DB	3.9	4.1	3.7	6.1	9.8	12.7	14.8	15.9	12.8	9.7	7.0	4.8
		MCWB	1.9	2.4	1.5	3.2	6.5	9.2	11.8	12.9	10.4	7.8	4.8	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	5.8	5.1	8.4	10.4	13.3	15.8	16.6	15.0	11.0	9.8	7.3
		MCDB	7.0	7.8	7.3	11.6	14.6	17.0	19.8	20.0	17.8	12.9	11.5	10.4
	2%	WB	4.2	4.8	3.8	5.7	8.5	11.8	14.2	15.3	12.8	10.0	7.9	5.2
		MCDB	6.2	6.7	6.0	8.2	12.7	15.3	17.2	17.9	15.0	11.9	9.6	7.4
	5%	WB	3.2	3.6	2.8	4.5	7.6	10.6	13.1	14.1	11.6	9.0	6.3	4.3
		MCDB	4.7	5.5	4.7	7.0	10.9	13.7	15.7	16.3	13.5	10.8	7.9	6.2
	10%	WB	2.4	2.7	1.7	3.5	6.7	9.7	12.3	13.1	10.7	7.9	4.9	3.1
		MCDB	3.8	4.4	3.5	5.9	9.4	12.5	14.2	15.3	12.5	9.4	6.8	4.7
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.0	6.3	6.2	6.6	6.4	5.5	5.4	5.7	5.7	5.5	5.9
		MCDBR	5.2	6.0	6.7	7.8	9.8	9.8	8.5	8.3	7.7	6.5	5.7	5.7
		MCWBR	4.3	4.7	5.2	5.7	6.3	6.1	5.4	5.3	5.3	4.9	4.8	4.6
	5% WB	MCDBR	4.9	5.8	6.6	7.5	8.6	8.7	7.7	7.6	6.7	6.1	5.9	5.7
		MCWBR	4.2	4.7	5.2	5.5	5.7	5.6	5.1	5.1	5.0	4.9	5.2	4.7
Clear-Sky Solar Irradiance	taub	0.290	0.334	0.354	0.389	0.401	0.395	0.379	0.379	0.358	0.331	0.318	0.279	
	taud	2.193	2.192	2.133	2.142	2.203	2.314	2.425	2.415	2.418	2.349	2.137	2.192	
	Ebn at Noon	575	696	784	818	834	846	854	828	790	703	512	483	
	Edh at Noon	53	82	112	130	130	119	104	99	85	69	56	41	
All-Sky Solar Radiation	RadAvg	0.42	0.96	2.34	3.62	4.56	4.89	4.24	3.51	2.46	1.32	0.54	0.29	
	RadStd	0.06	0.12	0.35	0.33	0.46	0.41	0.41	0.36	0.24	0.10	0.06	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page